

Oil Price Shocks and Civil Conflict: Evidence from Nigeria

Arinze Nwokolo 

Abstract

When and for what reason do governments choose to monopolize violence and consolidate power? Theory suggests three channels: when the government has coercive power against the opposition, if it shifts the distribution of power in its favor, and when contingent spoils are large. Using international oil price shocks and a novel dataset on oil-producing local government areas, this article examines how commodity prices affect civil conflict in Nigeria. Results show that a rise in oil price leads to a more than proportionate increase in government attacks on rebel groups in the oil region. The findings are consistent with the theoretical predictions: positive oil price shocks increase the monopoly of violence by the government through an increase in coercion, a rise in regaining territories from rebel groups, and an increase of violence in areas with large oil fields.

JEL classification: C23, D74, J30, L70, Q34

Keywords: oil price, natural resource, conflict, firms

1. Introduction

Contention over natural resources, between government and armed rebel groups, is a recurrent theme in many civil conflicts.¹ It remains unclear why governments choose to peacefully accommodate insurgents by buying them off or to consolidate power and monopolize violence by defeating them militarily. Theoretically, this puzzle is explained in three ways (Powell 2013). First, the government consolidates power and weakens the opposition civilly whenever it has coercive power. This mechanism occurs when the incumbent seeks to consolidate its position by taking over the internal security forces and arming its militia while weakening the opposition by arresting, eliminating, or isolating its leaders. The

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1 In the last six decades, no less than 40 percent of intra-state conflicts have been associated with natural resources (see, e.g., Lacina and Gleditsch 2005). Between 1997 and 2014, 320 of these events occurred in Sub-Saharan Africa (Raleigh et al. 2010).

incumbent can also achieve this either by repressing the opposition or by reducing its access to state power or increase transfers today though it is unable to commit to future transfers. Second, the presence of contingent spoils (such as the discovery of giant oil fields) influences a government's decision to monopolize violence. Contingent spoils are rents from natural resources that accrue to the government once rebels are defeated. These returns come from increased investment and economic activity by firms as a result of a higher level of security and protection in areas with natural resources. In other words, the higher the expected returns from contingent rent, the more conflict. Third, large shifts in the distribution of power between governments and rebel groups create commitment problems that lead to conflict. A shift in the distribution of power can occur through retaking territory from the opposition, and postponing or canceling local elections when the opposition is likely to win.

Other competing views from the literature show that the increase in resource rents makes conflict feasible by providing incentives for insurgents to engage in conflict,² and the decline in local wages reduces the opportunity cost of engaging in violent activities.³ Although large bodies of the literature study these channels at the cross-country level, the evidence for these mechanisms at the micro-level remains scarce. A plausible reason for this is the paucity of data on natural resource location at the subnational level in developing countries (Nillesen and Bulte 2014).⁴

To examine these mechanisms, this article uses firm-level data on oil-producing fields and wells to construct a 19-year annual panel of oil-producing local government areas. Specifically, it investigates the impact of an exogenous oil price shock on civil conflict by using data on conflict events and oil production in 774 local government areas in Nigeria between 1998 and 2016. The results show that a rise in global oil price increases conflict in oil-producing local government areas through more violence against civilians by rebel groups and the government, government attacks on rebel groups, and rebel-group attacks on government. The empirical analysis combines an original dataset from public records that documents yearly production data of oil firms and conflict events from the Armed Conflict Location Events Data (ACLED) (Raleigh et al. 2010). Following Berman, Shapiro, and Felter (2011), the article characterizes conflict events as a three-way interaction between rebels, the government, and civilians as follows: (a) rebel-group attacks on civilians, (b) rebel-group attacks on government, (c) government attacks on rebel groups, and (d) government attacks on civilians. These definitions capture the two distinct features of the civil conflict in the country. The first is the two-sided violence between rebel groups and the government. The second is the one-sided violence against civilians by rebel groups and the government. Using this data, this article interacts the international oil price with local government areas that produce oil to estimate the impact of price shock on local conflict. A possible confounder to the estimation is the concentration of oil production in the southern part of the country. This within-country heterogeneity may bias the analysis if the pre-sample local government area characteristics associated with the dynamics of conflict make the oil-producing and non-oil-producing local government areas unbalanced.⁵ The inclusion of local government area characteristics by year fixed effects, and local government area fixed effects control for this possibility. The positive change in oil price captures within-local-government-area

2 See Collier and Hoeffler (2004), Fearon (2005), Besley and Persson (2008, 2011).

3 See, for example, Miguel, Satyanath, and Sergenti (2004), Brückner and Ciccone (2010), Besley and Persson (2010), Collier and Hoeffler (1998). For a review of this literature see Blattman and Miguel (2010).

4 Recent studies show the importance of geographical concentration of natural resources for conflict. See Abadie and Gardeazabal (2003), Angrist and Kugler (2008), Bellows and Miguel (2009), Dube and Vargas (2013), Lujala (2010), Morelli and Rohner (2015). Caselli, Morelli, and Rohner (2015) show that the presence and location of oil are important predictors of interstate conflict.

5 For instance, the geographical terrain and educational attainment. Fenske and Zurimendi (2017) study the effect of an oil price shock on inequality between ethnic groups in Nigeria. They find that high oil prices increase schooling for southern ethnic groups. The interaction of district characteristics with year fixed effect is also used in other studies to control for omitted bias. See Berman, Shapiro, and Felter (2011).

variation in conflict events conditional on these covariates. Using non-oil local government areas as a counterfactual, the article shows that oil price shock differentially increases conflict in local government areas that produce oil.⁶

A 1-standard-deviation rise in the price of oil in the international market translates to a 5 percentage point increase in the intensity of rebel-group attacks on civilians, a 1 percentage point increase in rebel-group attacks on government, an 8 percentage point increase in government attacks on rebel groups, and a 2 percentage point increase in government attacks on civilians in oil-producing local government areas, relative to non-oil-producing local government areas. Consistent with the coercive power mechanism, this article shows that an increase in oil prices raised the incidents of government coercion in oil-producing local government areas. Similarly, it finds that oil-producing local government areas with giant oil discoveries are likely to experience more civil conflicts for any oil price increase. The result is consistent with the contingent spoil mechanism. Finally, it shows that a rise in oil price leads to an increase in the government's attempt to retake territories from rebel groups in oil-producing areas. Besides, oil price shock also reduces the probability of elections of local government council officials but increases the likelihood of the appointment of caretaker committees in oil-producing areas. These results are consistent with a shift in the power distribution mechanism. In contrast, there is no effect of oil price shock on revenue allocation or labor outcomes of households in these areas.

The article contributes to the current literature in several ways. First, it shows the importance of coercive power and contingent spoils as an incentive for governments to monopolize violence and consolidate power because they expect higher contingent returns associated with investments for the extraction of the natural resource.⁷ The findings relate to the recent study by [Sanchez de la Sierra \(2020\)](#), which shows that violent actors impose monopoly violence to extract taxes from miners. An interesting relationship with this article is the link between monopolizing violence and state formation, which is consistent with the theoretical underpinning of [Tilly \(1985\)](#). Besides, [Sanchez de la Sierra \(2020\)](#) shows that the monopoly is related to the increase in the global demand for coltan and persists years after a decline in demand. This effect is consistent with the findings, in this article, that violence persists even after a fall in oil prices.

Second, this article provides micro-level evidence of the importance of location effects. Onshore oil facilities are more vulnerable to attacks by rebel groups than offshore oil facilities. This observation is consistent with the contingent spoils mechanism by [Powell \(2013\)](#), in addition to studies by [Lujala \(2010\)](#) and [Ross \(2012\)](#). In particular, it is consistent with the literature on spatial conflict model analysis ([Adhvaryu et al. 2021](#); [Aragón and Rud 2013](#); [Berman et al. 2017](#); [Maystadt et al. 2013](#); [Sanchez de la Sierra 2020](#)) which use disaggregated data and similar econometric analysis. [Berman et al. \(2017\)](#) show that a rise in international mineral prices increases conflict risk in African countries that export primary commodities. In addition, the article shows that the characteristics of oil firms that extract oil in local areas exacerbate conflict and provide an incentive for insurgents to engage in criminal activities such as oil theft, the kidnapping of oil workers, and extortion.

Third, in contrast to the resource rent mechanism, the article shows that variation in oil price does not significantly affect revenue streams, nor does it increase federal transfers to oil-producing areas. These results are in line with the studies by [Fetzer and Kyburz \(2018\)](#) and [Cotet and Tsui \(2013\)](#). It differs from studies that show that oil revenue, and not oil location (source), increases the likelihood of conflict ([Collier and Hoeffler 2004](#); [Humphreys 2005](#)).⁸ Besides, using data on local elections, it finds that variation in oil

6 A similar approach is used by [Dube and Vargas \(2013\)](#).

7 It also relates to the recent research by [Lei and Michaels \(2014\)](#) which shows that the discovery of oil fields increases the incidence of internal armed conflict.

8 On theoretical models which show that rents from resources increase incentives for violence, see [Bates, Greif, and Singh \(2002\)](#), [Bates \(2008\)](#), [Besley and Persson \(2010, 2011\)](#), [Caselli and Michaels \(2013\)](#), and [Grossman \(1995\)](#). Other empirical studies on this topic include [Brollo et al. \(2013\)](#), [Vicente \(2010\)](#), and [Caselli and Michaels \(2013\)](#).

price reduces the probability of having an elected local government chairman and increases the likelihood of the appointment of caretaker committees in oil-producing local government areas.

Fourth, the article shows that oil shocks do not affect individual labor outcomes in resource-rich regions. This finding is consistent with predictions by Grossman (1991), Chassang and Padró i Miquel (2009), and Hirshleifer (1995) that a negative shock to the labor-intensive rather than the capital-intensive sector enhances the opportunity cost of rebellion. Finally, the article also relates to studies that show a relationship between negative income shocks and an increase in civil conflict (Collier and Hoeffler 1998; Dube and Vargas 2013; McGuirk and Burke 2020; Miguel, Satyanath, and Sergenti 2004). The article by Abidoye and Calì (2015) shows, on the one hand, that an increase in the price of consumed agricultural items reduced real income and the opportunity cost of violence for households in Nigeria and, on the other hand, that a rise in oil price increased conflict in oil-producing areas, but the violence decreased after the amnesty to rebel groups in 2009. In contrast, this article finds that civil conflict in oil-producing areas persists after the amnesty and after including controls for the variation of world price on exported agricultural items.

The rest of the article is structured as follows. In Institutional Setting we briefly review the oil industry and local conflicts in Nigeria, and we also present a theoretical model regarding consolidating power and monopolizing violence. Next we describe the data, outline the empirical strategy, and present results. Finally, we discuss possible mechanisms and then give conclusions.

2. Institutional Setting

Oil Industry and Local Conflicts

Oil exploration in Nigeria dates to 1908 with the search for oil deposits in the southwestern region of the country. The first oil was discovered in 1956 in Oloibiri, located in the Niger Delta region, and crude exports began in 1958. In 1961, total exports were dominated by cocoa, groundnut, and rubber with crude oil contributing 7.1 percent to the total export revenue. Between 1965 and 1970 the percentage share of crude oil to export earnings increased from 13.5 percent to 63.9 percent to become the leading source of foreign exchange (Obaje 2009). By 1979 it contributed 95 percent of total external earnings and generated 75 percent of government revenue. The strategic importance of crude oil to the Nigerian economy makes it vulnerable to international oil price volatility.

Currently, Nigeria has an estimated 37 billion barrels of proved crude oil reserves and 180 trillion cubic feet (Tcf) of proved natural gas reserves, mostly situated along the country's Niger Delta and offshore in the Bight of Benin, the Gulf of Guinea, and the Bight of Bonny.⁹ Commercial oil production is mainly in the Niger Delta region situated at the apex of the Gulf of Guinea on the African west coast. The region consists of nine oil-producing states with over 250 oil-producing communities and an extensive network of wells and production-related facilities.¹⁰ Endowed with huge oil and gas fields, half of which are offshore, the region produces over a million barrels of oil per day.

The oil industry operates under a statutory monopoly over mineral exploitation by the Nigerian government and regulated through the Nigerian National Petroleum Cooperation (NNPC). The NNPC operates through joint ventures and production-sharing contracts with oil majors. These firms receive territorial concessions (blocs) to extract oil.¹¹ Oil revenues are distributed to states through a derivation formula with a higher share to oil-producing states and communities.¹² However, intermittent changes to

9 See *Oil & Gas Journal* (2014).

10 The states are Abia, Akwa Ibom, Bayelsa, Cross River, Delta, Edo, Ondo, Imo, and Rivers.

11 The upstream sector is largely dominated by multinational exploration and production companies such as Royal Dutch Shell, Total Fina Elf, ExxonMobil, ENI/Agip, ChevronTexaco, and Addax Petroleum.

12 Oil-producing states currently receive 13 percent of revenue from oil receipts.

this revenue allocation strategy make it a source of tension in the Niger Delta region leading to demands for an increase in the amount of derivation or outright control of the natural resource.¹³

The Niger Delta disputes over resource control started in the early 1960s with the disruption of oil production through protests by the Movement for the Survival of the Ogoni People (MOSOP). By the late 1990s, conflicts intensified between rebel groups and the government due to grievances related to environmental and developmental neglect.¹⁴ Specifically, the rebel groups demanded greater local control and more transparent management of oil revenues, as well as adequate compensation of local communities for negative externalities derived from oil exploitation. By 2005, violent community conflicts in the Rivers, Bayelsa, and Delta states numbered 120–150 per year, and more than fifty armed groups with an estimated 20,000–25,000 armed youths were operating in the oil-producing region (UNDP 2006). The formation of the Movement for the Emancipation of the Niger Delta (MEND) escalated the conflict in 2006. The most coherent and trained armed group in the region, and estimated to have 5,000–10,000 combatants, MEND claimed responsibility for kidnapping oil workers and attacking onshore and offshore oil facilities while generating income for arms through the oil bunkering trade (Asuni 2009).¹⁵ In 2009, more than 20,000 ex-combatants accepted amnesty from the government and participated in a program of disarmament, demobilization, reorientation, and reintegration (DDRR) (Francis, Lapin, and Rossiasco 2011).

Empirical Question

Weak Institutions and Armed Conflict. Countries with weak institutions and resource abundance often face one or more armed rebel groups that challenge the government's control over certain areas of the country and the spoils that come with it. The government must decide how to deal with the opposition. First, it can attempt to accommodate or contain the rebel group through peace resolution without fighting. Second, it can try to monopolize violence and consolidate power by defeating them militarily or disarming them peacefully through a buyout. This attempt will lead to a large shift in the distribution of power against the rebel group, which will leave it in a weaker position for future bargains. The rebel group, in turn, can fight to hinder government efforts and stop or slow the adverse shift in the distribution of power, or it can accept the offer from the government and submit to its peaceful consolidation of power (Powell 2012).

Persistent Fighting and Negotiated Settlement. Since persistent fighting can prevent an adverse shift in power, the government and rebel groups face a trade-off. They can either fight now and risk being defeated, or they can avoid the immediate costs of fighting by calling a truce at the expense of being in a weaker position in the future. Nevertheless, fighting sometimes leads to a negotiated settlement, which in turn often breaks down into renewed fighting.

Recurrent Fighting. Armed conflict recurs because the government and rebel groups are unable to commit to future actions. On the one hand, the government cannot guarantee future transfers or cease future efforts to weaken the rebel group. On the other hand, the rebel group cannot commit not to fight in

13 The allocation formula of oil revenues has changed 18 times since 1946. See Ross (2003).

14 For example, the 1997 protest by 10,000 youths at the Alebiri to end the activities of Shell in the local government area and the 1998–99 mobilization of the Ijaw from the Ijaw Youth and National Council led to conflict with government forces and deepened the political disorder across the region. See Watts (2004).

15 It is estimated that between 70,000 and 300,000 barrels per day (more than 12 percent of the daily average oil production) are lost to illegal oil trade. For instance, Nigeria lost 136 million barrels of oil with an estimated value of \$11 billion to oil theft and sabotage between 2009 and 2011 (see Kent 2013). Further estimates show a loss of 84.8 million barrels at a cost of \$6.7 billion to oil theft in 2013 (Wallis 2015).

the future to prevent government consolidation. Hence, negotiated settlements break down in renewed fighting due to commitment problems.

Empirical Question. When and why do governments choose to monopolize violence and consolidate power? When does this lead to costly fighting rather than efforts to buy out the rebel group and thereby avoid considerable losses due to fighting (Powell 2013)? I present the theoretical model in the following subsection.

The Model

This section presents, in words, a simple model derived from Powell (2013). Consider a two-player infinite stochastic game in which two factions, the government and an armed rival faction (the rebel group), vie for state control and the spoils that come with it. Specifically, the factions try to divide a flow of “pies” in a state with weak institutions and rule of law. Owing to the institutional weakness of the state, neither faction can commit to how the pies will be divided in the future. In other words, whatever is agreed today can be renegotiated in the future, taking into consideration any change in the distribution of power. Hence, each faction is unable to commit to not fighting in the future and thereby remains armed and capable of fighting. Thus, each tries to maximize its sum of discounted spoils. This model leads to certain testable implications. First, in equilibrium, the government always consolidates power and monopolizes violence by weakening the rebel groups whenever it has “coercive power.” This consolidation shifts the distribution of power in its favor through the takeover and politicizing of the internal security forces, arming its militia, arresting, eliminating, or isolating the leaders of rebel groups. Therefore, coercive power creates an incentive for the government to consolidate power and monopolize violence. Second, although peaceful consolidation avoids the deadweight loss of fighting, it takes time to buy the rebels off, weaken them, and ultimately eliminate them. Since the government cannot commit to future transfers, it faces a liquidity problem because it is unable to offer the rebels enough today for being weaker tomorrow. This commitment problem limits the rate at which peaceful consolidation can occur and delays the realization of gains from contingent spoils. If the government lacks coercive power and is unable to consolidate power, then contingent spoils ensure that the government monopolizes violence either by fully compensating the rebels today or by eliminating them as peacefully as possible. Third, the size of the contingent spoils creates the key trade-off that leads to fighting. In other words, when fighting occurs is determined by the trade-off between the cost of delaying the contingent spoils and the gain of fighting on better terms later. The smaller the contingent spoils, the more the cost of fighting outweighs the cost of delay, and the government monopolizes violence peacefully. By contrast, the larger the contingent spoils, the higher the opportunity cost of delay, and the more likely the government is to consolidate power through fighting.

3. Data

The research design exploits the fact that conflict intensity within oil-producing local government areas depends on oil price changes in the world market and group competition for resource rents. Naturally, to study this we require data on oil production within the local economy, oil prices, conflict events amongst different groups (both government and rebel groups), local-government-area-level income, and state revenues. The summary statistics are in table 1.

Conflict Data. The article uses conflict data recorded by the Armed Conflict Location and Event Data Project (ACLED). The data covers all countries in Sub-Saharan Africa from 1997–2018. The data contains real-time reports on daily violent and non-violent events such as battles, riots, protests, violence against civilians by political actors, including rebels, governments, communal groups. The empirical analysis focuses on conflict events in Nigeria between 1998 and 2016, a 19-year window that captures conflict

Table 1. Summary Statistics

	Observation	Mean	Standard deviation	Minimum	Maximum
<i>Panel A: Conflict intensity outcomes</i>					
Rebel group attacks civilians	14,706	0.0255	0.421	0	28
Rebel group attacks government	14,706	0.0422	0.0975	0	7
Government attacks rebel group	14,706	0.0409	0.710	0	43
Government attacks civilians	14,706	0.0186	0.185	0	6
Rebel group attacks rebel group	14,706	0.000272	0.0165	0	1
Battle—Government regains territory	14,706	0.0130	0.254	0	18
Battle—Non-state actor overtakes territory	14,706	0.00184	0.0553	0	3
<i>Panel B: Conflict incidence outcomes</i>					
Rebel group attacks civilians	14,706	0.00898	0.0943	0	1
Rebel group attacks government	14,706	0.00292	0.0540	0	1
Government attacks rebel group	14,706	0.0112	0.105	0	1
Government attacks civilians	14,706	0.0139	0.117	0	1
Rebel group attacks rebel group	14,706	0.000272	0.0165	0	1
<i>Panel C: Commodity prices</i>					
Log oil price, millions of 2017 naira per barrel	14,706	4.449	0.543	2.444	4.957
<i>Panel D: Demographics</i>					
Percentage of households with primary school education in 1990	14,706	5.357	14.26	0	96.60
Elevation (meters)	14,706	268.0	219.5	1	1,494
<i>Panel E: Federal transfer, ethnic political power, coercive power, local election</i>					
Log district revenue, millions of 2017 naira	13,158	20.55	0.856	18.355	22.75
Ethnic group lost power in previous year	14,706	0.0599	0.237	0	1
Local government chairman election	14,706	0.202	0.402	0	1
Caretaker committee appointment	14,706	0.701	0.458	0	1
Government coercive power	14,706	0.789	6.015	0	209
Rebel kidnapping	14,706	0.296	3.979	0	232
<i>Panel F: Household Labor outcomes</i>					
Log real wage (2006–2016)	96,571	9.155	1.172	0.693	16.99
Log labor hours (2006–2016)	96,571	4.968	0.326	0.693	5.075

Source: Author's analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, the Demographic Health Survey, NASA's SRTM3, Federal Accounts Allocation Committee (FAAC) reports, the Ethnic Power Relations (EPR) Dataset, local elections in Nigeria from Factiva, the Integrated Crisis Early Warning System (ICEWS), and the Nigeria General Household Survey (NGHS) of 2006–2016.

Note: Panel A and B variables are from ACLED. The panel C variable is from the WTI series. Panel D variables are from the Demographic Health Survey. Panel E variables are from FAAC reports, EPR, Factiva, ICEWS, and NGHS.

trends. Events are observed at the local government area level over time using data specific information on the date, location, event type, geographic coordinates, and contextual notes. To capture the local violence cycle and distinguish who attacks, conflict actors are aggregated into groups defined as rebel-group attacks on civilians, rebel-group attacks on government, government attacks on rebel groups, and government attacks on civilians. Each dependent variable is constructed using the interaction code that highlights the relationship between two conflict actors. Specifically, for each violent interaction between the three groups, the article uses the total number of events to capture conflict intensity and binary variables that show whether the events occurred within each year of the study period.

Oil Location Data. To identify oil-producing local government areas, firm-level information on local-government-area oil production is collated from the Nigerian National Petroleum Corporation (NNPC) annual reports. To complement this, additional information from secondary sources such as annual

reports of oil firms and concession maps to locate exploration local government areas were used. The full dataset shows an average annual number of 155 oil wells of more than 1,700 million barrels in production in 46 oil-producing local government areas across nine states over the sample period.¹⁶

Oil Price Data. The oil price measure is the average annual spot oil price from the West Texas Intermediate (WTI) series.¹⁷ The variation in global oil prices can be safely assumed to be exogenous since Nigeria produces less than 3 percent of world oil supply.¹⁸ The study focuses primarily on the period 1998–2016 because it contains oil shocks of sign and magnitude comparable to those of the oil boom in the 1970s. The oil price rose in 1998 from \$14.39 per barrel to a high of \$99.57 per barrel in 2008 before it fell to \$61.69 per barrel in 2009.¹⁹ The substantial variation in oil price levels over this period means that it is possible to make inferences about changes in violent events in Nigeria by comparing oil price levels within a relatively narrow time window.

Data on Mechanisms. Using data from the Integrated Crisis Early Warning System (ICEWS) (Boschee et al. 2015), coercive power and rebel extortion mechanisms are constructed from the number of events regarding government coercive power and rebel kidnapping in local government areas in Nigeria between 1998 and 2016. ICEWS is an event dataset of coded interactions between sociopolitical actors. It contains more than 17 million georeferenced events from 1995 to 2018 for more than 198 countries. Each event identifies a source actor, an event type (using the Conflict and Mediation Event Observations (CAMEO) taxonomy of events), and a target actor. Contingent spoils analysis uses the Horn (2003) dataset on Giant Oil and Gas Fields of the World, which spans the period from 1868 to 2004. Using this dataset, the Nigerian local government areas with giant oil fields before 1998 are interacted with the log of oil price.

The mechanisms on the share of the distribution of power are from three distinct datasets. The mechanism of “government regains territory” is from the ACLED data set and concerns the government recapture of territories. This variable is the number of battle events between government and rebels, whereby the government regained territories previously lost to rebel groups. The “ethnic group loses power” mechanism is from the Ethnic Power-Relations Dataset (Vogt et al. 2015). This dataset identifies 733 politically relevant ethnic groups in over 155 countries between 1946 and 2016. For every country it shows the executive powers attained by these ethnic groups. Using this dataset, a dummy variable is constructed that captures whether an ethnic group has lost power in the previous year. For mechanisms related to local government chairman election and caretaker committee appointment, following Fetzer and Kyburz (2018), a media content analysis of Nigerian news articles from Factiva is used to construct a dataset on local government council elections and the appointment of caretaker committees between 1998 and 2016 in 774 local government areas in Nigeria. For each local government, two binary variables are defined: a binary variable equal to 1 if a local government council official was elected and another binary variable equal to 1 if the state governor selected a caretaker committee.

16 The oil-producing states are Abia, Akwa Ibom, Bayelsa, Cross River, Delta, Edo, Imo, Ondo, and Rivers. There have also been recent oil discoveries in Anambra and Lagos.

17 The choice of price index relies on the fact that the United States has traditionally been the largest importer of Nigerian oil, until 2012. India is currently the largest importer of Nigerian crude oil at 370,000 barrels per day (bbl/d) while the United States is the 10th largest importer at 60,000 bbl/d. Europe remains the largest regional importer of Nigerian oil at 900,000 bbl/d. See United States Energy Information Administration (2015). The oil price data is available at <https://fred.stlouisfed.org/series/WTISPLC>.

18 See <https://www.eia.gov/international/data/world/>. In addition, Griffin (1985) show that although Nigeria is part of OPEC, it does not coordinate its production quantities with OPEC because it is a small producer with competitive fringe tendencies.

19 At the nominal price, it was at an all-time high of \$145 per barrel on July 3, 2008.

Mechanisms on opportunity costs are constructed using labor outcomes data from household surveys. Specifically, data on wage and employment are from the Nigeria General Household Survey of 2006–2016.²⁰ The surveys provide information on age, sex, marital status, local government area, wages, hours worked, employment, and migrant status. The sample includes all persons born in 774 local government areas that are at least 15 and not more than 65 years old between 1998 and 2016.²¹ Using this panel, a monthly real wage and hours employed is calculated to estimate the opportunity cost of civil conflict.²² To study the impact of revenue allocation, data on revenue allocation of oil revenues is collated from transfers between the federal government and local government areas. Intergovernmental transfers from the Federation Account are overwhelming the primary source of revenue for subnational governments in Nigeria.²³ The central government shares oil and VAT revenues with the 36 states (excluding the Federal Capital Territory) and the 774 local governments (local government areas) of the country. According to the current revenue allocation formula, 52.68 percent of the transfers are allocated to the Federal government, 26.72 percent to state governments, and 20.6 percent to local governments.²⁴ The fact that revenue from oil represents a substantial share of total public sector income means that the sharing of oil revenue dominates intergovernmental relations in Nigeria and is also a potential source of appropriation. The Office of the Account General of the Federation publishes the revenue allocated to the federal, state, and local governments every month.²⁵ Using this data, monthly allocations are aggregated to generate annual allocation figures at the local government area level. The online appendix provides additional details on the data characteristics.

4. Empirical Strategy and Results

Empirical Specification

To motivate our story, [fig. 1](#) shows a graphical display of the international oil price movement and our conflict outcomes in Nigeria. The graph reveals the correlation between oil price movement and conflict events. Differentiating the conflict events between oil and non-oil local government areas, [fig. 2](#) shows that the increase in global oil price relates to more violence in oil local government areas. This growth is particularly evident for events during the 2002–2008 oil boom. The local government areas with oil production data at the beginning of the sample period are shown in [fig. 3](#). The map shows the oil-producing local government areas to be mostly in the southeastern part of the country. Not all local government areas in the south, however, produce oil.²⁶

The empirical specification follows a difference-in-difference estimation with the exogenous variation from the annual oil price movement in the international market. The oil location variable is a dummy variable for local government areas that produced oil in 1998. Using oil location in the first year of the sample ensures that conflict events over the analysis period do not correlate with local oil supplies or oil

20 Unfortunately, there is no district-level data for the Nigeria General Household Survey from 1995 to 2005.

21 The minimum age for work in Nigeria is 12 years.

22 Although the same individual is not observed over time, the availability of the surveys allows the observation of the average wage and employment within the specified age groups over time. A similar method is used by [Dube and Vargas \(2013\)](#).

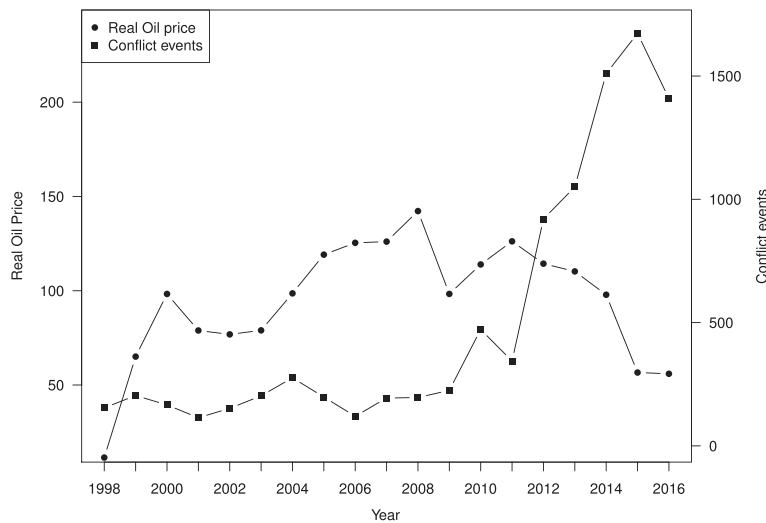
23 Internal revenues of most states are below 10 percent of their total revenues.

24 For states that produce oil, the amount each state receives varies according to the number of local governments within the state and the amount of oil produced (oil-producing states receive an additional 13 percent from the Federation Account).

25 This is available at the Office of the Accountant-General of the Federation website from 2007. See <http://www.oagf.gov.ng/>.

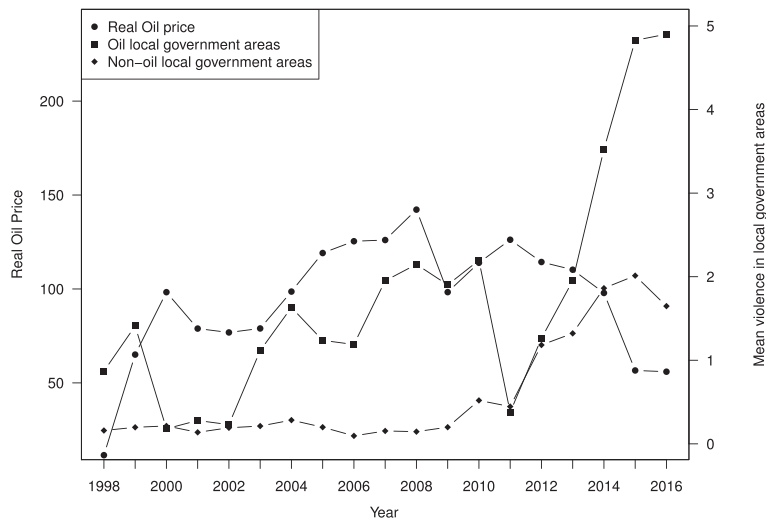
26 There are no oil local government areas in the north and there is a huge economic disparity between the north and south. The poverty rate in the north is twice that of the southern region. The northern region accounts for the majority (66 percent) of the poor in the country. See [World Bank \(2014\)](#).

Figure 1. Oil Price and Conflict Outcomes



Source: Oil price data is from the West Texas Intermediate (WTI) and conflict data is from the Armed Conflict Location Events Data (ACLED).
Note: Real oil price is in Nigerian naira and conflict event is the total number of violent events, per year, between the government and rebel groups.

Figure 2. Oil Price and Violence in Local Government Areas



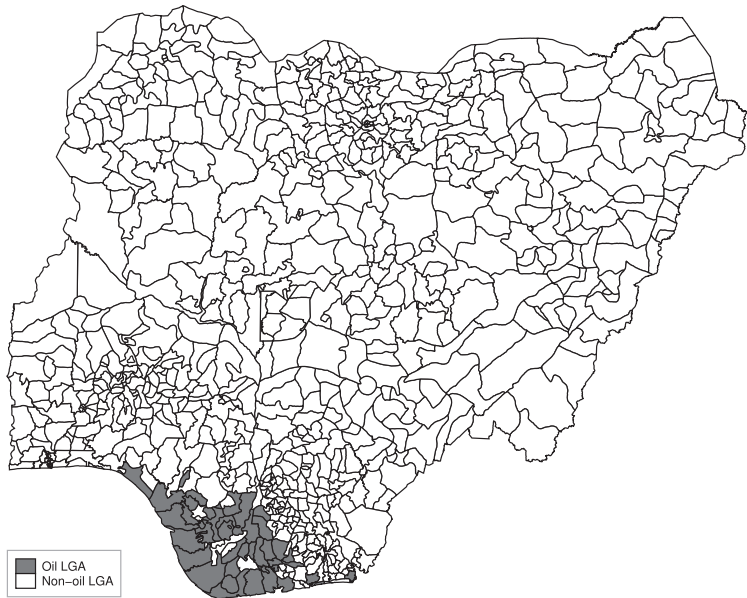
Source: Oil price data is from the West Texas Intermediate (WTI) and conflict data is from the Armed Conflict Location Events Data (ACLED).
Note: Real oil price is in Nigerian naira. Mean violence in local government areas is the average number of violent events, per year, between the government and rebel groups in oil and non-oil local government areas.

field discoveries. The main results of the article are from the following model:

$$Y_{jrt} = \alpha_1(Oil_j \times OilPrice_t) + \alpha_2 X_{jrt} + \delta_j + \gamma_t + \lambda_{rt} + \epsilon_{jrt}, \tag{1}$$

where Y_{jrt} are conflict outcomes comprising rebel-group attacks on civilians, rebel-group attacks on government, government attacks on rebel groups, and government attacks on civilians in local government area j , region r , and year t . In the main specification, the different conflict outcomes Y_{jrt} regress on $(Oil_j \times OilPrice_t)$, a variable that captures the interaction between local government areas that produced

Figure 3. Oil Local Government Areas in 1998



Source: Oil local government area (LGA) data is constructed from the annual statistical bulletin of the Nigerian National Petroleum Corporation (NNPC) from 1998 to 2016.

Note: Oil LGA are local government areas that produced oil in 1998 while non-oil LGA are local government areas without oil production in 1998.

oil in 1998 (a dummy variable equal to 1, and 0 otherwise) and the exogenous oil price in the international market (stated in natural log terms to capture the percentage change effect of oil price), a vector of pre-sample covariates interacted with time X_{jrt} , local government area fixed effects δ_j , year fixed effects γ_t , regional linear trends λ_{rt} , and an error term ϵ_{jrt} .

The vector of covariates X_{jrt} includes pre-sample indicator variables such as the percentage of households with primary school education and geographical elevation. These variables capture characteristics between oil and non-oil local government areas before our sample. As shown in table S1 in the supplementary online appendix, there are no significant variations in population, average years of schooling, percentage of households with secondary education, the proportion of literate households, and different measures of household access to public services. Nevertheless, there are differences between oil and non-oil areas as measured by the percentage of households with primary education and the geographic elevation, which can confound our estimation. These variables, in addition to the regional linear trends λ_{rt} , are included in the analysis as controls. Equation (1) is estimated using ordinary least squares (OLS) for the different conflict events. Limiting the sample to these events is important because it captures the violence cycle and helps to disentangle the effect of oil from the effect of other time-varying factors that influence conflict in Nigeria. The baseline analysis focuses on comparing the effect of an oil price increase between oil-producing and non-oil-producing local government areas. A potential concern in using OLS is that it produces a biased estimate of α_1 because pre-sample characteristics may be correlated with time and thus with $(Oil_i \times OilPrice_t)$. Using a difference-in-differences (DD) analysis addresses these confounding factors. The identifying assumption is that, conditional on the covariates, the average conflict events for the oil and non-oil local government areas would have followed a parallel trend in the absence of oil production.²⁷ Thus α_1 captures the average effect of oil price on conflict outcomes in oil-producing

27 This is related to the model proposed by Abadie (2005).

local government areas. The standard errors are clustered at the local government area level. Spatial and temporal autocorrelations are accounted for by using the methods of [Conley \(1999\)](#) as implemented by [Hsiang, Meng, and Cane \(2011\)](#). The specified distance for the spatial dimension is 100 km, while the time horizon is three years.²⁸

Main Results

The estimates of the baseline specification in [equation \(1\)](#) are in [table 2](#). For each conflict outcome, a simple “naive” version of the estimation model is reported and controls subsequently added to capture the pre-sample difference in local government area characteristics. Columns (1)–(8) show results of the measures of conflict outcome: rebel-group attacks on civilians, rebel-group attacks on government, government attacks on rebel groups, and government attacks on civilians by the number of events. For the four conflict outcomes, an increase in the price of oil is associated with higher levels of civil conflict intensity and are significant at the 1 percent level. The results for all estimates remain the same after the inclusion of local government area covariates by year fixed effects. The estimated effects are large and dissimilar in magnitude. In columns (1) and (2), the positive coefficient on $(Oil_i \times OilPrice_t)$ shows that a spike in the price of oil in the global market increases the rebel-group attacks on civilians. The result implies that if the price of oil increases by 1 standard deviation, the probability of violence against civilians by rebel groups (relative to the mean) will increase by 185 percent.²⁹ For rebel-group attacks on government (columns 3 and 4), the results show that a standard deviation rise in oil price increases this event by 11 percent of the mean. Estimates for government attacks on rebel groups are in columns (5) and (6). The coefficient of interest, 0.140, implies that a standard deviation increase in the price of oil raises the degree of government attack on rebel groups by 186 percent of the mean. Columns (7) and (8) consider the probability that the government might attack civilians. The coefficient shows a 96 percent increase, relative to the mean, for a 1 standard deviation increase in oil price. An additional estimate of the impact of an oil price shock on conflict incidences are in [table 3](#). All conflict outcomes are significant, and the magnitudes of the effects are substantial. Relative to the mean, the results show that a standard deviation rise in oil price will increase the frequency of rebel-group attacks on civilians by 236 percent (columns 1 and 2), rebel-group attacks on the government by 167 percent (columns 3 and 4), government attacks on rebel groups by 238 percent (columns 5 and 6), and government attacks on civilians by 66 percent (columns 7 and 8).

Robustness

This section considers the possibility that the baseline estimates are likely biased by issues related to the definition of oil local government areas, oil price, oil production intensity, the dataset on violence and conflict characteristics, measurement error, or omitted variables.

Alternative Definition of Oil Local Government Areas

The baseline specification of $(Oil_i \times OilPrice_t)$ in [equation \(1\)](#) is a natural estimation to address the endogenous concern of opening and closing of oil fields. The use of oil local government areas at the beginning of the sample period (1998) ensures that the exogeneity comes from demand shock in the international oil price market. However, this specification assumes that conflict events in these local government areas do not influence oil production activity. Following [Berman et al. \(2017\)](#), three strategies are used to address this concern. First, the estimation of [equation \(1\)](#) is restricted to local government areas that always produced oil from 1998 to 2016. This estimation ensures that the coefficient of interest, α_1 ,

28 As reported in the supplementary online appendix, the baseline results are robust to alternative specifications of spatial and local-government-area-specific correlation.

29 The point estimate is larger in magnitude than the effect found in [Dube and Vargas \(2013\)](#) with respect to paramilitary attacks.

Table 2. Oil Price Shock and Civil Conflict Intensity

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Rebel group attacks civilians	Rebel group attacks civilians	Rebel group attacks civilians	Rebel group attacks government	Government attacks rebel groups	Government attacks rebel groups	Government attacks civilians	Government attacks civilians
Oil LGA × log oil price	0.087*** (0.023) [0.023]	0.087*** (0.023) [0.024]	0.013*** (0.005) [0.005]	0.013*** (0.005) [0.005]	0.140*** (0.029) [0.038]	0.140*** (0.030) [0.038]	0.032*** (0.020) [0.015]	0.033*** (0.020) [0.015]
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LGA covariates × year	No	Yes	No	Yes	No	Yes	No	Yes
Regional linear time trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,706	14,706	14,706	14,706	14,706	14,706	14,706	14,706

Source: Author's analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, and the Demographic Health Survey. Note: Each column represents a separate regression. Oil price shock is the interaction between global oil prices (in log terms) and local government areas (LGAs) that produced oil in 1998. Conflict outcomes capture the number of times each violent event occurred. For all regressions, robust standard errors clustered at LGA level are in parentheses and Conley standard errors computed at a 100km cutoff are in brackets. The sample period is 1998–2016. The LGA-level covariates include percentage of households with primary school education in 1990 and LGA elevation (in meters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Oil Price Shock and Civil Conflict Incidence

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Rebel group attacks civilians	Rebel group attacks civilians	Rebel group attacks civilians	Rebel group attacks government	Government attacks rebel groups	Government attacks rebel groups	Government attacks rebel groups	Government attacks civilians
Oil LGA $\times$ log oil price	0.039*** (0.010) [0.010]	0.039*** (0.010) [0.010]	0.009*** (0.003) [0.003]	0.009*** (0.003) [0.003]	0.049*** (0.010) [0.014]	0.049*** (0.010) [0.014]	0.017** (0.008) [0.010]	0.017** (0.008) [0.010]
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LGA covariates $\times$ year	No	Yes	No	Yes	No	Yes	No	Yes
Regional linear time trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,706	14,706	14,706	14,706	14,706	14,706	14,706	14,706

Source: Author's analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, and the Demographic Health Survey.

Note: Each column represents a separate regression. Oil price shock is the interaction between global oil prices (in log terms) and local government areas (LGAs) that produced oil in 1998. Conflict outcomes are binary variables that capture whether each type of violent event occurred. For all regressions, robust standard errors clustered at LGA level are in parentheses and Conley standard errors computed at a 100km cutoff are in brackets. The sample period is 1998–2016. The LGA-level covariates include percentage of households with primary school education in 1990 and LGA (in meters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

captures the influence of oil price conditional on stable local production activity. Second, oil local government areas are defined as local government areas that ever-produced oil between 1998 and 2016. Third, oil prices are interacted with a lagged oil local government area dummy over the sample period to capture the fact that future conflicts might influence current production. The results for the different strategies are in table S2 in the supplementary online appendix. The coefficient of interest remains positive and significant for conflict events regarding government attacks on rebel groups and civilians in all specifications. Panel A shows results related to permanent oil production throughout the period. The coefficient and the magnitude are comparable to the baseline results.³⁰ The results in Panel B (oil local government area that ever produced between 1998 and 2016) are also similar to the results from baseline specification. The point estimates and magnitudes for lagged dummy variables for oil local government area (Panel C), for all conflict events, are larger than the estimates in the baseline. However, only government attacks on rebels and civilians show a significant effect. The robustness of the definition of oil local government areas is reexamined by using an alternative dataset that captures local government areas that produced oil before the sample period (i.e., before 1998–2016), such as the global and georeferenced petroleum dataset (PETRODATA) of oil and gas fields (Lujala, Rød, and Thieme 2007). This dataset consists of over 1,200 records of oil fields (known to exist) in 114 countries with information on the year of first discovery and initial production year.³¹ The years of discovery were 1956–1972, while the earliest year of production was between 1958 and 1978. Using this data, oil prices are separately interacted with the year of discovery and year of first oil production. Apart from government attacks on civilians, the results, as reported in table S3 in the supplementary online appendix, are comparable to the baseline estimates (with varying effect of magnitude) and also show a positive and significant relationship for rebel-group attacks on government and government attacks on rebel groups.³² The issue of endogeneity of oil-producing local government areas is also examined by instrumenting the local government area dummy in the sample period in the original dataset with its preceding equivalent from PETRODATA. The point estimates, as shown in table S4 in the supplementary online appendix, are larger and significant at a 1 percent level for conflict events related to rebel-group attacks on civilians and government attacks on rebel groups.³³

Alternative Definition of Oil Price

One of the assumptions in equation (1) is that oil price has an instantaneous effect on conflict. This premise is relaxed by introducing two separate time lags in the baseline model to capture the effect of the timing of oil price shocks on conflict outcomes. First, the conflict types are regressed by a one-year time lag ($Oil_i \times OilPrice_{t-1}$) and a one-year lead time ($Oil_i \times OilPrice_{t+1}$). The results are shown in table S5 in the supplementary online appendix. The point estimates for the two specifications are approximately close to the baseline results.³⁴ Second, this exercise is replicated by controlling for the first difference of a one-year lag and lead in prices and the results are in table S6 in the supplementary online appendix. The coefficients

30 In this specification, a standard deviation rise in oil price increases the probability of rebel-group attacks on civilians by 107 percent, rebel-group attacks on government by 10 percent, government attacks on rebel groups by 185 percent, and government attacks on civilians by 96 percent.

31 See Caselli, Morelli, and Rohner (2015) for a paper using a similar data combined with a conflict data set. The PETRODATA dataset can be accessed from <https://www.prio.org/Data/Geographical-and-Resource-Datasets/Petroleum-Dataset/>.

32 The magnitude of effect in panel A (year of discovery) ranges from a conflict probability increase of 162 percent for rebel-group attacks on civilians to 167 percent for government attacks on rebel groups. For panel B (year of initial production) this changes to 162 percent for an attack on civilians by rebel groups and 155 percent for an attack on rebel groups by the government.

33 A similar methodology is used in Berman et al. (2017).

34 Under government attack on civilians, the coefficient remains insignificant when the current price level is combined with a time lag of one year and a lead time of one year. However, the coefficient is stable and significant for other conflict events.

are larger and significant compared to the baseline specification. The positive and significant impact of future oil prices on rebel-group attacks on civilians may seem puzzling, as one would not expect armed rebel groups to anticipate future oil prices. Besides, if oil prices are highly correlated, it may be misleading to include leads as a placebo test.³⁵ There are two possible explanations for these results. First, bargaining between the government and rebel groups relies on how big or small the spoil is. A lack of transparency creates an information asymmetry regarding the size of the spoils. In other words, the government has a better idea of the size of the spoils than rebel groups. The rebel groups, however, know when oil prices are high or low and, therefore, whether the spoil is on average large or small. If the actual size of the spoils is small relative to the expected size, which both the government and rebel group know, then the probability of fighting is higher. The rebels can choose to increase the level of violence on civilians in order to maximize the expected violence costs imposed on the government.³⁶ Second, the expected future price of oil is an incentive and motivation for rebel groups to engage in lucrative criminal activities such as oil theft, extortion, kidnapping, and selling contraband or future oil rights to foreign investors (Ross 2012).

Oil Production Intensity

The geographical location of oil is a proxy for oil production. The use of within-country location(s) of oil may be an imperfect measure of oil extraction and production intensity. Oil production depends on other factors than conflict events. The production of oil in Nigeria is primarily carried out either as a joint venture (JV) company between the government and private firms or under a production sharing contract (PSC). Government participation in joint ventures ranged from 55 to 60 percent, and the management of this venture is through a subsidiary of NNPC called the National Petroleum Investment Management Service (NAPIMS).³⁷ Foreign oil companies that operate under PSC do not pay royalties until production offsets the initial investment outlay. Subsequently, they share the cost value of production and royalties with the government. The impact of oil production is considered in two ways. First, by interacting oil prices with oil production in 1998 the different production variations in oil local government areas at the beginning of the sample period are captured. Second, by distinguishing the two margins of production through interacting oil price, with average oil production between 1998 and 2016 (the intensive margin) and with the average number of oil-producing wells in the sample period (the extensive margin). In panel A of table S7 in the supplementary online appendix (using oil production in 1998), the results are broadly similar in point estimate and magnitude to the baseline results. In panel B, the specification using oil production between 1998 and 2016 instead of oil location in 1998 also has a positive, significant and larger coefficient compared to the baseline estimate of table 2 (0.125 vs. 0.087) for rebel-group attacks on civilians (column 1) and (0.017 vs. 0.013) for rebel-group attacks on government (column 2) and (0.173 vs. 0.140) for government attacks on rebel groups (column 3). A standard deviation increase in oil price hence translates to an increase in magnitude (relative to the mean) of conflict probability from 185 to 266 percent for violence against civilians by rebel groups and from 186 to 230 percent for government attacks on rebel groups. The use of oil-producing wells, in panel C, shows similar results with a substantial increase in magnitude. Compared to the baseline estimates, the size of the magnitude increased by 130 percentage points for rebel-group attacks on civilians and by 110 percentage points for government attacks on rebel groups. Finally, other robustness checks are performed and reported in the supplementary online appendix. They are briefly listed as follows: (a) alternative dataset on violence and conflict characteristics (tables S8 and S9), (b) measurement error (table S10), (c) omitted variables (tables S11 to S14), (d) conflict spillover to neighboring non-oil-producing local government areas (tables S15 and S16), (e) two-way clustering by geographic and time structure of oil prices (tables S17 to S19),

35 On the discussion of this issue, see McGuirk and Burke (2020).

36 For a theoretical model regarding spoils politics, see Dal Bó and Powell (2009).

37 All joint ventures have 60 percent government participation. An exception to this is the joint venture with Royal Dutch Shell which is at 55 percent. See International Monetary Fund (2004).

(f) Poisson fixed effects (table S20), (g) controlling for state by year fixed effects (table S21), (h) interacting the price of other commodities with oil production (table S22), (i) ascertaining the persistence of violence (table S23), (j) controlling for the effect of food prices on civil conflict (table S24), (k) controlling for the effect of the revenue allocation formula (table S25), (l) estimating the effect of oil price on civil conflict after the 2009 amnesty (table S26), (m) estimating the impact of oil price on civil conflict in southern local government areas (table S27), (n) restricting the time frame to the democratic period (table S28), (o) controlling for region-specific year effects (table S29), (p) restricting the sample to 2000–2016 (table S30) and to 2006–2016 (table S31), (q) controlling for potential endogeneity of location of oil production (table S32) and other conflict outcomes (table S33).

5. Discussions

This section examines the possible explanations for the likelihood of conflict increase due to positive oil price shocks. Understanding the influence of oil price on the decision of the government to accommodate rebel groups or monopolize violence by rebel disarmament is important in explaining the conflict level results, as well as for assessing the extent to which the experience in Nigeria can be generalized. Overall, the evidence indicates that, relative to non-oil local government areas, positive oil price shocks increase the government's use of coercive power against the opposition in oil-producing local government areas. Furthermore, government would monopolize violence in local government areas with sizable contingent spoils (large oil fields) and try to alter the distribution of power in these areas as a result of oil price shock. In contrast, the results show no price effect either on revenue allocation or on individual wages or labor hours. An important contributory factor to the high level of oil-related conflict is the role of oil firms in producing local government areas. Previous studies on the relationship between conflict and firms have usually focused on the impact of conflict on stock market returns of firms ([Abadie and Gardeazabal 2003](#)), stock market reaction to conflict events ([Guidolin and La Ferrara 2007](#)), or the response of firms to violence ([Ksoll, Macchiavello, and Morjaria 2016](#)). In this section, the effects of firm characteristics on conflict outcomes are studied. The idea is to examine how the physical presence of oil firms affects the probability of conflict in local government areas. As shown later, the ownership, size, and the country of the firm headquarters have a substantial impact on conflict events.

Coercive Power and Shift in the Distribution of Power

The model, described in the section Institutional Setting, states that oil prices can affect conflict by making the government consolidate power by exploiting its coercive power and shifting the distribution of power against the opposition. In this subsection, these channels are investigated by analyzing whether changes in oil price affect the number of events relating to government coercion, and battles that lead to regaining territories from rebels and whether oil shocks affect incidents of loss of power by relevant ethnic groups, the election of local government council officials, or the appointment of caretaker committees. The results of the impact of oil price on government use of coercive power are presented in column (1) of [table 4](#). The estimates imply that a standard deviation increase in oil price increases the intensity of government coercion by 20 percent of the mean or roughly 3 percentage points in standard deviations. Next, the mechanism of the shift in the distribution of power is examined. The result in column (2) shows that oil price shock significantly increases battles that lead to government recapture of territories, on average, by 146 percent in local government areas that produce oil. In contrast, the coefficient in column (3) shows that the price shock reduces the incidences of the relevant ethnic group losing political power in the area by 82 percent. To investigate why this is the case, the effect of oil prices on incidents of local government elections are analyzed. The results in columns (4) and (5) show that, relative to the mean, a rise in oil price will reduce the election of local government council officials by 11 percent and increase the appointment

Table 4. Coercive Power and Shift in Distribution of Power Mechanisms

Dependent variable	Coercive power	Shift in distribution of power			
	(1) Government coercion	(2) Government regains territory	(3) Ethnic group loses power	(4) Local chairman election	(5) Caretaker committee appointment
Oil LGA × log oil price	0.289* (0.174) [0.170]	0.035*** (0.008) [0.010]	−0.090*** (0.008) [0.027]	−0.040*** (0.009) [0.030]	0.129*** (0.013) [0.041]
Year fixed effects	Yes	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes	Yes
LGA covariates × year	Yes	Yes	Yes	Yes	Yes
Regional linear time trends	Yes	Yes	Yes	Yes	Yes
Observations	14,706	14,706	14,706	14,706	14,706
Sample period	1998–2016	1998–2016	1998–2016	1998–2016	1998–2016

Source: Author’s analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, the Demographic Health Survey, the Integrated Crisis Early Warning System (ICEWS), and Factiva.

Note: Each column represents a separate regression. Oil price shock is the interaction between global oil prices (in log terms) and local government areas (LGAs) that produced oil in 1998. Government coercion is the number of events where the government used coercive tactics against the opposition. Government regains territory are the number of violent events related to battles between the government and rebel groups by which geographical areas are recovered. Ethnic group loses power is a binary variable that indicates whether a politically relevant ethnic group lost power in the previous year. Local chairman election is a binary value that captures whether local elections occurred in LGAs. Caretaker committee appointment is a dummy variable that shows the selection of custodians of the LGAs by the state government. For all regressions, robust standard errors clustered at LGA level are in parentheses and Conley standard errors computed at a 100 km cutoff are in brackets. The LGA-level covariates include percentage of households with primary school education in 1990 and LGA (in meters). Each conflict event captures the annual number of attacks or clashes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of caretaker committees by 10 percent. These estimates imply that though the government might militarily defeat the rebel groups, it is unable to weaken its access to local political power.

Contingent Spoils

To examine why government and rebel groups renege on cease-fire agreements and engage in cost of fighting, this subsection analyzes the possible role that contingent spoils play in influencing government decisions to monopolize violence. As mentioned earlier, these spoils encompass the gains the government reaps from providing additional commitment, in terms of security, to oil exploration and production firms. According to the theoretical model of Powell (2013), contingent spoils generate a trade-off that can lead to fighting. The government will eliminate the opposition as peacefully as possible if they are small but will monopolize violence by fighting if they are large. This is investigated by slightly adjusting the specification in equation (1):

$$Y_{jrt} = \alpha_1(GiantOil_j \times OilPrice_t) + \alpha_2(Oil_j \times OilPrice_t) + \alpha_3X_{jrt} + \delta_j + \gamma_t + \lambda_{rt} + \epsilon_{jrt}, \tag{2}$$

where $(GiantOil_j \times OilPrice_t)$ is an indicator variable that captures the interaction between local government areas with giant oil discoveries before or in 1998 (a dummy variable equal to 1, and 0 otherwise) and the exogenous oil price increase in the international market (stated in natural log terms to capture the percentage change effect of oil price). The results for the conflict outcomes are shown in columns (1) to (4) of table 5. The estimates indicate that oil price shock exerts a substantial effect on civil conflict in local governments with large oil discoveries. The coefficients imply that, on average, an increase of 1 standard deviation in oil price increases government attacks on rebel groups by 184 percent in local government areas with large oil discovery, compared to a 153 percent increase in government attacks on rebel groups in local government areas with smaller oil discoveries.

Table 5. Contingent Spoils Mechanism

Dependent variable	Contingent spoils			
	(1) Rebel group attacks civilians	(2) Rebel group attacks government	(3) Government attacks rebel groups	(4) Government attacks civilians
Giant oil LGA × log oil price	0.0922 (0.0828) [0.0632]	0.0063 (0.0091) [0.0082]	0.1386** (0.0586) [0.0513]	0.0180 (0.0177) [0.0193]
Oil LGA × log oil price	0.0697** (0.0299) [0.0224]	0.0122** (0.0054) [0.0054]	0.1150*** (0.0297) [0.0327]	0.0293*** (0.0088) [0.1508]
Year fixed effects	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes
LGA covariates × year	Yes	Yes	Yes	Yes
Regional linear time trends	Yes	Yes	Yes	Yes
Observations	14,706	14,706	14,706	14,706
Sample period	1998–2016	1998–2016	1998–2016	1998–2016

Source: Author's analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, the Demographic Health Survey, and the dataset on Giant Oil and Gas Fields of the World.

Note: Each column represents a separate regression. Giant Oil LGA × log oil price is the interaction between global oil prices (in log terms) and local government areas (LGAs) with giant oil discoveries prior to or in 1998. Oil price shock (Oil LGA × log oil price) is the interaction between global oil prices (in log terms) and LGAs that produced oil in 1998. Conflict outcomes capture the number of times each violent event occurred. For all regressions, robust standard errors clustered at LGA level are in parentheses and Conley standard errors computed at a 100 km cutoff are in brackets. The LGA-level covariates include percentage of households with primary school education in 1990 and LGA (in meters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Revenue Allocation and Household Income

Next, the impact of a positive oil price shock on revenue allocation to local government areas and household income are considered. Federal transfers to local government areas come from the Federation Account. This account receives payments regarding oil revenues, company income tax proceeds, customs duties, and excise taxes. Oil revenues represent more than 70 percent of total revenue from the Federation Account.³⁸ Before intergovernmental transfers, the Federation Account is subject to initial deductions known as first charges. These first charges are deducted from oil revenues in the Federation Account and include a 13 percent allocation of oil revenue to the oil-producing states.³⁹ The remaining amount is shared amongst the federal, state, and local governments according to a derivation formula. The effect of oil price on federal transfers is assessed by using the transfer of oil revenue from federal to local government areas as the dependent variable. The results in column (1) of table 6 show a small and insignificant effect of oil price shock on fiscal transfers to local government areas that produce oil. To explore whether a rise in oil price motivates rebel groups to engage in criminal activities such as kidnapping and extortion, data on rebel kidnapping from ICEWS is used. The results show that an oil price increase of 1 standard deviation increases kidnapping by 140 percent in the mean oil-producing local government area.

The impact of oil price on household labor hours and wages is examined through the specification

$$W_{jrt} = \alpha_1(Oil_j \times OilPrice_t) + \alpha_2 P_{jrt} + \delta_j + \tau_t + \lambda_{rt} + v_{jrt}, \quad (3)$$

38 Revenue from oil is generated from the sale of crude oil and gas: signature bonuses, royalties, and petroleum profit tax (PPT) with a rate of 85 percent (65.75 percent in the first five years of production).

39 Other charges are comprised of external debt service, the share of government production cost of oil, the cost of government-sponsored projects, and National Judiciary Council expenditure.

Table 6. Revenue Allocation, Rebel Extortion and Opportunity Cost Mechanisms

Dependent variable	Revenue allocation	Rebel extortion	Opportunity costs	
	(1) Federal transfers	(2) Rebel kidnapping	(3) Log labor hours	(4) Log wages
Oil LGA $\times$ log oil price	0.005 (0.008) [0.011]	0.763* (0.421) [0.281]	−0.024 (0.043) [0.046]	0.0091 (0.142) [0.097]
Year fixed effects	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes
LGA covariates $\times$ year	Yes	Yes	Yes	Yes
Regional linear time trends	Yes	Yes	Yes	Yes
Household controls	No	No	Yes	Yes
Observations	13,158	14,706	96,751	96,751
Sample period	2000–2016	1998–2016	2006–2016	2006–2016

Source: Author's analysis based on data from Federal Accounts Allocation Committee (FAAC) reports, the Integrated Crisis Early Warning System (ICEWS), and the Nigeria General Household Survey (NGHS) of 2006–2016.

Note: Each column represents a separate regression. Oil LGA $\times$ log oil price is the interaction between global oil prices (in log terms) and local government areas (LGAs) that produced oil in 1998. Conflict outcomes capture the number of times each violent event occurred. For all regressions, robust standard errors clustered at LGA level are in parentheses and Conley standard errors computed at a 100 km cutoff are in brackets. Federal transfers is the natural log of annual intergovernmental transfers of oil revenue from the federal government to the local government. The LGA-level covariates include percentage of households with primary school education in 1990 and LGA (in meters). Household controls include educational level attained, age, age squared, rural status, gender, and marital status. Log labor hours is the log of hours worked in the last month and log wages is the log of hourly wages defined as individual earnings per hour worked in the last month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

where W_{irt} is the log of hourly wage and hours worked of individual i in local government area j , region r , and at time t . The vector of covariates P_{irt} includes educational attainment, age, age squared, rural status, gender, and marital status. The right-hand side also includes local government area fixed effects δ_j , year fixed effects τ_t , regional linear trends λ_{rt} , and error term v_{irt} . The results in columns (3) and (4) of [table 6](#) show the point estimates of labor hours and wages to be small and insignificant.⁴⁰

Firm Characteristics

The results shown so far have made minimal use of the information on firm characteristics. However, firm characteristics may explain the circumstances that motivate rebel groups and the government to engage in conflict. As explained in the section Institutional Setting, the key features of the Nigerian civil conflict are comprised of the tendency of rebel groups to disrupt oil production and the ability of government to suppress these groups. The nature of the conflict, therefore, implies that oil firms can either pay rebel groups to refrain from attacks (or pay ransom for kidnapped employees)⁴¹ or elicit a protective response from the government by paying fewer royalties due to a reduction in production.⁴² According to [Ross \(2012\)](#), oil firms are unusually prone to extortion from rebel groups for three reasons. The first is the location of oil and the willingness of oil firms to enter hazardous regions. Firms in this industry must follow the oil even if it leads them to politically unstable and conflict-prone areas such as the Niger Delta in Nigeria. In other words, their ability to work in places with high-security risks implies that they will have a close interaction with rebels determined to raise money. Second, oil exploration and production involve huge investments in fixed assets. This commitment gives oil firms a strong incentive to stay put

40 This can be a consequence of the capital-intensiveness of oil production or the small labor share of oil in Nigerian employment.

41 For instance, in March 1997 over 100 Shell workers were taken hostage by Ijaws armed with automatic rifles, and oil installations were occupied by groups of Ijaws. See [Frynas \(2000\)](#).

42 This invariably means less revenue for the government.

to recoup their initial investment and make huge profits from mature facilities, hence their willingness to reach an agreement with the government or rebel groups to protect their facilities. Third, oil firms earn enormous rents from their activities. Consequently, they can afford the security costs that come from working in risky environments, such as paying large sums for kidnapped employees or making payments to rebels to refrain from attacks. In addition, these events compound the hostility towards foreign firms established or closely associated with colonial rule.

These hypotheses are studied using a feature of the dataset that captures the firm identity and the oil fields operated by each oil firm at the beginning of the sample period. A strategy similar to [Berman et al. \(2017\)](#) is followed by (i) differentiating firms by ownership (foreign firms, domestic public and domestic private firms), by colonial origin (whether the firm has colonial ties to Nigeria), and by firm size (whether it is a major oil firm in Nigeria) and (ii) interacting the share of oil production of each firm category (at the beginning of the sample period) within each oil-producing local government area with the coefficient of interest ($Oil_j \times OilPrice_t$). Of the firms, 98 percent are foreign-owned, 1 percent are owned by the government, and the remaining 1 percent are private-owned. In this period, 60 percent of oil production is from foreign companies with headquarters in the former colonizing country (the United Kingdom).⁴³ Empirically, the specification is the following:

$$\begin{aligned} Y_{jrt} = & \alpha_1(Oil_j \times OilPrice_t) + \alpha_2(Oil_j \times FirmCategory_j) \\ & + \alpha_3(OilPrice_t \times FirmCategory_j) \\ & + \alpha_4(Oil_j \times OilPrice_t \times FirmCategory_j) \\ & + \alpha_5 X_{jrt} + \delta_j + \gamma_t + \lambda_{rt} + \epsilon_{jrt}, \end{aligned} \quad (4)$$

where $(Oil_j \times OilPrice_t \times FirmCategory_j)$ is the interaction between local government areas that produced oil in 1998, the exogenous oil price in the international market (stated in natural log terms to capture the percentage change effect of oil price), and the exclusive category of firms that produced oil in 1998 in local government area j (captured by the share of the total oil produced by firm type). Firms are differentiated by ownership (foreign firms, domestic public firms, and domestic private firms), colonial ties, or whether the firm has headquarters in the United Kingdom, and firm size (if the firm is a major player in the Nigerian oil industry). Given that the three shares (foreign, public, and private firms) sum to 1 and to account for collinearity, the baseline interaction term $(Oil_j \times OilPrice_t)$ is dropped from the specification. The results are shown in columns (1)–(3) of [table 7](#). We see that the share of foreign and public (government-owned) firms increases the impact of oil price on rebel-group attacks on civilians, rebel-group attacks on government, and government attacks on rebel groups. In contrast, there is a negative and significant effect on private (indigenous) firms. These results are consistent with the idea that foreign and public firms are more likely to pay extortion money to rebel groups to avoid or cease attacks on production facilities.⁴⁴ In contrast, private indigenous firms, due to their size, are less “visible” and constitute a less attractive target. Besides, it shows that the government’s willingness to use military force and monopolize violence increases in local government areas with a large presence of foreign firms. This finding is consistent with the documentation of government crackdown in oil-producing communities by

43 The Nigerian oil industry is dominated by foreign- and government-owned companies such as Shell, Mobil, Chevron, Elf (Total), Agip, Texaco, and Nigerian Petroleum Development Company (NPDC). For instance, Shell Development Petroleum Company (SPDC) accounts for 43 percent of total oil output in Nigeria. It owns more than 1,000 oil-producing wells in the Niger Delta region. See [Catan and Mahtani \(2006\)](#). Examples of local private firms include Dubri Oil, Express Petroleum Oil & Gas, Allied Energy Resources, ConOil Producing, Platform Petroleum, Seplat, MidWestern Oil & Gas, Pillar Oil, and Atlas Petroleum.

44 An interesting event regarding this is the formation of a militant group by the name Isongoforo. This group was provided with payment and hired by foreign companies for protection purposes. See [Watts \(2004\)](#) for a detailed account of these issues.

Table 7. Firm Ownership

Dependent variable	(1) Rebel group attacks civilians	(2) Rebel group attacks government	(3) Government attacks rebel groups	(4) Government attacks civilians
Oil LGA × log oil price × foreign firms	0.090*** (0.024) [0.018]	0.014*** (0.005) [0.005]	0.145*** (0.031) [0.025]	0.035*** (0.010) [0.016]
Oil LGA × log oil price × public firms	0.036*** (0.016) [0.020]	0.005* (0.003) [0.003]	0.062** (0.027) [0.033]	−0.037*** (0.008) [0.078]
Oil LGA × log oil price × private firms	−0.395* (0.215) [0.284]	−0.060 (0.045) [0.058]	−0.594*** (0.224) [0.424]	−0.005 (0.093) [0.349]
Year fixed effects	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes
LGA covariate × year	Yes	Yes	Yes	Yes
Regional linear time trends	Yes	Yes	Yes	Yes
Observations	14,706	14,706	14,706	14,706

Source: Author’s analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, the Annual Statistical Bulletin of the Nigerian National Petroleum Corporation (NNPC), and the Demographic Health Survey.

Note: Each column represents a separate regression. Oil price shock is the interaction between global oil prices (in log terms) and local government areas (LGAs) that produced oil in 1998. For all regressions, robust standard errors clustered at the LGA level are in parentheses and Conley standard errors computed at a 100 km cutoff are in brackets. The sample period is 1998–2016. The LGA-level covariates include percentage of households with primary school education in 1990 and district elevation (in meters). Each conflict event captures the annual number of attacks or clashes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

a human rights group.⁴⁵ Another plausible explanation for this result is the close relationship between foreign oil firms and the government. As mentioned earlier, foreign oil firms usually form a joint venture with NNPC, the state-owned oil company. In addition, they have agreements with government agencies ranging from land acquisition to arms purchase for the security forces (Frynas 2001). The result is the same for the use of violent repression against civilians (column 4). However, there is a negative and significant effect regarding civilian suppression in local government areas with public oil firms. There is no significant effect of oil prices on districts with private firms in this regard. Finally, in order to examine the possibility that the colonial origin of the firms magnifies the impact of oil price shocks on conflict,⁴⁶ foreign firms are differentiated into colonizer or non-colonizer. The point estimates in table 8 show that foreign firms with colonial and non-colonial affiliations are associated with conflict related to rebel-group attacks on civilians (columns 1), rebel-group attacks on government (column 2), government attacks on rebel groups (columns 3), and government attacks on civilians (columns 4).⁴⁷ Furthermore, under the interaction with large firms category, the table shows that these conflicts increase in local government areas where the major oil firms operate.

45 See Human Rights Watch (1999b). In addition, it is reported that oil firms lent helicopters and boats to the government for an attack on ethnic groups in the Niger Delta (Human Rights Watch 1999a).

46 That firms with colonial ties receive more concessions from the governments of former colonies is supported by the literature. See Frynas, Beck, and Mellahi (2007), Stockwell (2004), White (2000).

47 These results are in contrast with those of Berman et al. (2017) who find no relationship between foreign firms with headquarters in colonial countries and conflict after mineral price shock.

Table 8. Colonial Origin of Firms and Firm Size

Dependent variable	(1) Rebel group attacks civilians	(2) Rebel group attacks government	(3) Government attacks rebel groups	(4) Government attacks civilians
Oil LGA $\times$ log oil price $\times$ colonizer	0.099*** (0.037) [0.026]	0.017** (0.008) [0.008]	0.161*** (0.041) [0.036]	0.039*** (0.013) [0.021]
Oil LGA $\times$ log oil price $\times$ non-colonizer	0.073*** (0.019) [0.017]	0.009** (0.004) [0.004]	0.116*** (0.029) [0.026]	0.029*** (0.010) [0.016]
Oil LGA $\times$ log oil price $\times$ large firms	0.090*** (0.024) [0.018]	0.014*** (0.005) [0.005]	0.145*** (0.031) [0.025]	0.035*** (0.010) [0.016]
Year fixed effects	Yes	Yes	Yes	Yes
LGA fixed effects	Yes	Yes	Yes	Yes
LGA covariate $\times$ year	Yes	Yes	Yes	Yes
Regional linear time trends	Yes	Yes	Yes	Yes
Observations	14,706	14,706	14,706	14,706

Source: Author's analysis based on data from the Armed Conflict Location and Event Data Project (ACLED), the West Texas Intermediate (WTI) series, the Annual Statistical Bulletin of the Nigerian National Petroleum Corporation (NNPC), and the Demographic Health Survey.

Note: Each column represents a separate regression. Oil price shock is the interaction between global oil prices (in log terms) and local government areas (LGAs) that produced oil in 1998. For all regressions, robust standard errors clustered at LGA level are in parentheses and Conley standard errors computed at a 100 km cutoff are in brackets. The sample period is 1998–2016. The LGA-level covariates include percentage of households with primary school education in 1990 and district elevation (in meters). Each conflict event captures the annual number of attacks or clashes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6. Conclusions

This article estimates the impact of oil price shock on civil conflict in Nigeria. It shows, using a novel dataset on oil-producing local government areas, that a high oil price increases the likelihood of rebel-group attacks on civilians, rebel-group attacks on government, and government attacks on rebel groups. The magnitude of these attacks, relative to the mean, ranges from 17 percent to over 180 percent between 1998 and 2016. The results are robust to alternative specifications such as using various definitions of oil-producing local government areas, oil price and production, different datasets on oil-producing local government areas and conflict events, and pre-sample conflict trends.

Consistent with the view that governments in weak states either accommodate insurgent groups or monopolize violence and consolidate power by disarming them (Powell 2013), the article shows that a rise in oil price by 1 standard deviation increases government coercion by 20 percent, shifts the distribution of power to the government by 142 percent, and intensifies government attacks in local government areas with large contingent spoils. There is no evidence that the oil price boom increases federal transfers to local governments that produce oil nor induces conflict through the opportunity cost channel.

Oil firms play a key role in civil conflict within oil-producing areas (Ross 2006, 2012). That is, the behavior and characteristics of oil firms are important both as a central feature and potential motivation for government and rebel groups to engage in violence. The article demonstrates that firm ownership, especially foreign- and government-owned firms, magnifies the probability of conflict as a result of oil price shock. Further evidence suggests that the colonial origin of firms also affects conflict. Firms with colonial ties are associated with more conflict probability than firms without such affiliations. Although previous work (mostly related to natural minerals in Africa; see Berman et al. 2017) shows that ownership and colonial origin of firms in mining regions can intensify conflict, this article is the first to consider these features in the context of oil extraction in Nigeria.

Overall, the results show the need for more attention to factors that influence the tendency of government, especially in Sub-Saharan Africa, to monopolize violence. In this regard, this article complements the literature on the role of conflict actors in state formation (Sanchez de la Sierra 2020; Tilly 1985). The tendency of governments to dominate the concentrated means of coercion, and when and under what circumstances this dominance is enforced, adds to the research on the causes of civil conflict and carries an important caution for policy makers in developing countries.

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